Session 7 — After-Session Checklist

ExInt II: Research Designs in SME Research · WU Vienna · SS 2026

Work through every step in order. Tick the box when done.

Reference project:
https://github.com/vkiefner/sme-intl/tree/main

DEADLINE: June 28, 2026 — 23:59

Where to work:
- VS Code
- Terminal
- Zotero
- GitHub
- Canvas

A References — Zotero, library.bib, and apa.csl

☐ Zotero

Step 1 — Check Zotero library covers theory, empirical, and methods

Every research note needs references in three areas. Verify 2-3 papers in each:

```
1. Theory — the theories your hypotheses are built on
```

```
2. Empirical — prior studies on your specific topic
```

```
3. Methods — justifying panel FE, clustered SEs, winsorizing
```

■ *Methods citations are often forgotten. You need at least one paper each for: panel fixed effects, clustered standard errors, and winsorizing.*

☐ Zotero, VS Code

Step 2 — Confirm library.bib is auto-exporting and citation keys are correct

Open references/library.bib and check the timestamp. Note the exact citation key for every paper you plan to cite — e.g. cohenAbsorptiveCapacityNew1990.

✎ **`references/library.bib`**

■ *If library.bib is not updating: Zotero → Edit → Preferences → Better BibTeX → verify Keep updated is enabled.*

☐ Terminal, GitHub

Step 3 — Download apa.csl and commit it to Git

Run once from the project root — this fixes the apa.csl not found error:

```
Invoke-WebRequest -Uri "https://raw.githubusercontent.com/citation-style-language/styles/master/apa.csl" -OutFile "references\apa.csl"
```

■ *Mac / Linux:*

```
curl -o references/apa.csl https://raw.githubusercontent.com/citation-style-language/styles/master/apa.csl
```

```
git add references/apa.csl
```

```
git commit -m "add apa.csl citation style"
```

```
git push
```

■ *apa.csl must be committed — the examiner's task all also needs it. It is a small XML file, safe to commit.*

B Taskfile and YAML — automate the PDF correctly

VS Code
Terminal

Step 4 — Rename research_note.md to research_note.qmd

Quarto requires the .qmd extension for documents with executable Python chunks:

```
Rename-Item research_note.md research_note.qmd
```

Mac / Linux:

```
mv research_note.md research_note.qmd
```

```
git add research_note.qmd
git rm research_note.md
git commit -m "rename to .qmd for executable chunks"
git push
```

VS Code

Step 5 — Fix Taskfile.yml — render standalone, all runs everything

render must have NO deps so task render does not trigger the full pipeline. all runs the analysis chain then renders:

Taskfile.yml

```
render:
desc: "Render research_note.qmd to PDF via Quarto"
cmds:
- quarto render research_note.qmd
all:
desc: "Full pipeline — analysis + PDF"
deps: [regression]
cmds:
- quarto render research_note.qmd
```

task render → PDF only. task all → pull → clean → descriptives → regression → PDF.

VS Code

Step 6 — Set up the YAML header in research_note.qmd

Complete YAML header — GitHub URL in author block appears on title page:

✎ **research_note.qmd**

```
---
title: "Your title"
author: |
Your Name | ExInt II | WU Vienna | SS 2026
https://github.com/[you]/[your-repo]
date: "June 2026"
bibliography: references/library.bib
csl: references/apa.csl
format:
pdf:
toc: false
number-sections: true
geometry: margin=2.5cm
fontsize: 11pt
linestretch: 1.3
execute:
eval: true
echo: false
warning: false
message: false
---
```

■ *eval: true runs Python chunks during rendering. echo: false hides the code. Make sure task all has run first so output/ files exist.*

Terminal

Step 7 — Install Quarto and tinytex if not already done

Install Quarto from quarto.org if not present, then run once:

```
quarto install tinytex
```

C Write the research note — section by section

VS Code

Step 8 — Setup chunk — load all output files at the top

Add a setup chunk immediately after the YAML header that loads everything once:

```
```{python}
#| label: setup
import pandas as pd, numpy as np
summary = pd.read_csv('output/tables/summary_statistics.csv', index_col=0)
results = pd.read_csv('output/tables/regression_results.csv', index_col=0)
panel = pd.read_parquet('data/processed/panel_with_vars.parquet')
n_obs = int(panel.shape[0])
n_firms = int(panel['gvkey'].nunique())
# compute any inline numbers you need here
```
```

■ *All variables set here are available throughout the document in inline {python} expressions and later chunks.*

VS Code

Step 9 — Introduction — motivation, gap, RQ (AMJ: set the hook)

Three paragraphs: what is known → what is NOT known (the gap) → what this note does. End with a block quote research question:

```
> **Research question:** [One sentence, ending with a question mark.]
```

■ *Do not describe a field — state a gap. The gap is what the cited literature has NOT answered.*

VS Code

Step 10 — Theory — build each hypothesis from theory (AMJ: ground your H)

Theory → mechanism → prediction → formal hypothesis. Each H in a block quote:

```
> **H1:** [Directional claim in one sentence.]
> *(Test: expected sign of coefficient)*
```

■ *Cite at least 2-3 papers per hypothesis. If your H is obvious without the mechanism argument, you have not built a case.*

VS Code

Step 11 — Data — use inline Python for sample N, no placeholders

Use inline expressions to pull N directly from the parquet — never type numbers manually:

```
The sample comprises `{python} f"{n_obs:,}"` firm-years from
`{python} f"{n_firms:,}"` unique firms.
```

Include: sample filters applied, date range, variable table with field names and formulas, regression equation in LaTeX, FE justified, clustered SEs cited, winsorizing mentioned.

VS Code

Step 12 — Results — automated table + inline coefficients

Regression table via Python chunk — reads directly from CSV:

```
```{python}
#| label: tbl-regression
#| tbl-cap: "Regression results. * p<0.10, ** p<0.05, *** p<0.01."
results[['(1) OLS','(2) TWFE','(3) TWFE+H2']]
```
```

Insert figure — path relative to project root:

~~~
![Caption.](output/figures/main_relationship.png){width=90%}
~~~

State each hypothesis as SUPPORTED or NOT SUPPORTED. Compare OLS vs FE and state the % difference.

■ *AMJ principle: every coefficient needs one plain-language sentence. Do not just say 'as shown in Table 2'.*

VS Code

Step 13 — Discussion and Conclusion (AMJ: broaden the return)

Discussion: link result back to theory. Conclusion: answer the RQ directly, two specific limitations, one future direction.

```
Final line — References section:
# References
::: {#refs}
:::
```

D Run task all and verify the PDF

Terminal

Step 14 — Run the full pipeline

```
.venv\Scripts\activate
task all
```

■ *Mac / Linux:*

```
source .venv/bin/activate && task all
```

Runs: pull → clean → descriptives → regression → quarto render research_note.qmd → research_note.pdf

■ *If pull times out, run python code/02_clean.py directly. The render step only needs output/ files to exist.*

VS Code

Step 15 — Open research_note.pdf and verify every element

```
Title page: name + GitHub URL visible
Sections: numbered automatically (1, 2, 3...)
Citations: (Author, Year) format — not [@key]
Table: regression results present, all three models
Figure: main_relationship.png visible
Inline N: actual numbers, not [N] placeholders
References: full list at end, only cited papers
```

Terminal
VS Code

Step 16 — Fix common errors

```
.qmd extension error -> rename .md to .qmd
apa.csl not found -> Step 3: download and commit
bibliography not found -> check references/library.bib path
LaTeX error -> quarto install tinytex
Citation shows [@key] -> key mismatch, check library.bib
ModuleNotFoundError -> venv not active when rendering
task render runs all -> remove deps from render task
```

E Commit, push, and submit

VS Code

Step 17 — Verify .gitignore

```
data/raw/
data/processed/
.env
research_note.pdf
*.log
*.tex
```

■ *The PDF is submitted via Canvas. apa.csl IS committed — needed for reproducibility.*

Terminal
GitHub

Step 18 — Final commit and push

```
git add code/ README.md research_note.qmd
git add references/library.bib references/apa.csl
git add Taskfile.yml requirements.txt pyproject.toml .env.example
git commit -m "final submission"
git push
```

Browser
GitHub

Step 19 — Verify repo on GitHub

Open your repo and confirm — data/ must NOT appear:

```
code/*.py research_note.qmd references/library.bib
references/apa.csl Taskfile.yml requirements.txt
pyproject.toml .env.example README.md
```

Canvas

Step 20 — Submit PDF via Canvas

Canvas → ExInt II → Assignments → Research Note Submission → upload research_note.pdf. Open the PDF first and confirm your GitHub repo URL is visible on the title page.

■ *Deadline: June 28, 2026 — 23:59.*

Terminal

Step 21 — Voluntary peer reproducibility check

Find a classmate on the same OS, clone their repo, run task all:

```
git clone https://github.com/[classmate]/[repo]
cd [repo]
python -m venv .venv && .venv\Scripts\activate
pip install -r requirements.txt
# add WRDS credentials to .env
task all
```

■ *Same OS is important. Single best way to catch problems before the deadline.*

F Final checklist — before June 28, 2026

- Zotero library has theory, empirical AND methods papers
- references/library.bib up to date — all cited papers present
- references/apa.csl downloaded, saved, and committed to Git
- File renamed: research_note.qmd (not .md)
- Taskfile: render task has NO deps — standalone
- Taskfile: all task has deps:[regression] and cmds: quarto render
- YAML: bibliography and csl paths correct
- YAML: GitHub URL in author field (title page)
- YAML: eval: true, echo: false, warning: false
- Setup chunk loads all output files at the top
- Introduction: RQ stated explicitly in a block quote
- Theory: logic chain theory -> mechanism -> prediction -> H
- Theory: each H cited with 2-3 papers, expected sign stated
- Data: N loaded from parquet via Python — no placeholder
- Data: variable table with field names, formulas, roles
- Method: LaTeX equation, FE justified, clustered SEs cited
- Results: table rendered from CSV via Python chunk
- Results: figure inserted from output/figures/
- Results: each H stated as SUPPORTED or NOT SUPPORTED
- Results: OLS vs FE % difference discussed
- Discussion: result linked back to cited theory
- Conclusion: RQ answered, two limitations, future direction
- References section with ::: {#refs} ::: block

- ☐ task all completes and produces research_note.pdf
- ☐ PDF: title page shows name + GitHub URL
- ☐ PDF: sections numbered, citations formatted, figure visible
- ☐ PDF: inline numbers match table (both from same output files)
- ☐ .gitignore: data/, .env, research_note.pdf NOT in repo
- ☐ apa.csl committed to references/
- ☐ All scripts, research_note.qmd, library.bib pushed
- ☐ research_note.pdf submitted via Canvas before June 28 at 23:59

Examiner workflow: clone repo → python -m venv .venv → pip install -r requirements.txt → add WRDS credentials to .env → task all → open research_note.pdf. Assessment: GitHub reproducibility 25% · Results quality 35% · Method 20% · Hypothesis development 15% · Originality 5%. Deadline: June 28, 2026 — 23:59.